

tcs2

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1 DFA

Let $A = (Q, \Sigma, \delta, q_0, F)$ be a DFA. Let x be a string. Let $\hat{\delta} : Q \times \Sigma^* \rightarrow Q$ be defined by $\hat{\delta}(q, \epsilon) = q$ and $\hat{\delta}(q, xa) = \delta(\hat{\delta}(q, x), a)$ for all $q \in Q, x \in \Sigma^*$.

Let $q \in Q, x, y \in \Sigma^*$. I claim that $\hat{\delta}(q, xy) = \hat{\delta}(\hat{\delta}(q, x), y)$.

Proof: I induct on $|y|$. If $|y| = 0$, then $y = \epsilon$, for which

$$\hat{\delta}(\hat{\delta}(q, x), \epsilon) = \hat{\delta}(q, x) = \delta(\hat{\delta}(q, x), \epsilon) = \hat{\delta}(q, x\epsilon).$$

Now suppose $z \in \Sigma^*, |z| = n + 1$, and the result holds for all $y \in \Sigma^*$ with $|y| \leq n$. Write $z = ya$ for some $a \in \Sigma$. Then

$$\begin{aligned} \hat{\delta}(\hat{\delta}(q, x), z) &= \hat{\delta}(\hat{\delta}(q, x), ya) = \delta(\hat{\delta}(\hat{\delta}(q, x), y), a) = \delta(\hat{\delta}(q, xy), a) \\ &= \hat{\delta}(q, (xy)a) = \hat{\delta}(q, x(ya)) = \hat{\delta}(q, xz). \end{aligned}$$

Note that the associativity of concatenation was used.

Let $q \in Q, x \in \Sigma^*$ and $a \in \Sigma$. I claim that $\hat{\delta}(q, ax) = \hat{\delta}(\delta(q, a), x)$.

Proof: Let $\hat{a}$ denote the string with one symbol a . By the previous result, $\hat{\delta}(\hat{\delta}(q, \hat{a}), x) = \hat{\delta}(q, \hat{a}x)$. Note that $\hat{a}x = ax$ and $\hat{\delta}(q, \hat{a}) = \delta(\hat{\delta}(q, \epsilon), a) = \delta(q, a)$, so $\hat{\delta}(\delta(q, a), x) = \hat{\delta}(q, ax)$, as desired.

2 ϵ -NFA

Let $E = (Q, \Sigma, \delta, q_0, F)$ be an ϵ -NFA. Sps $R \subset Q$ and define $\text{Cl}_n(R)$ by $\text{Cl}_0(R) = R$ and $\text{Cl}_{n+1}(R) = \bigcup_{q \in \text{Cl}_n(R)} \delta(q, \epsilon)$. Let $\text{Cl}(R) = \bigcup_{n < \omega} \text{Cl}_n(R)$.

Let $\{R_i\}_{i \in I}$ be a collection of subsets of Q and $R = \bigcup_{i \in I} R_i$. I claim that $\text{Cl}_n(R) = \bigcup_{i \in I} \text{Cl}_n(R_i)$.

Proof: I induct on n . $n = 0$ is obvious. Sps the result for n . Then

$$\begin{aligned}\text{Cl}_{n+1}(R) &= \bigcup_{q \in \text{Cl}_n(R)} \delta(q, \epsilon) = \bigcup_{q \in \bigcup_{i \in I} \text{Cl}_n(R_i)} \delta(q, \epsilon) \\ &= \bigcup_{i \in I} \bigcup_{q \in \text{Cl}_n(R_i)} \delta(q, \epsilon) = \bigcup_{i \in I} \text{Cl}_{n+1}(R_i),\end{aligned}$$

completing the induction.

Sps $n, m < \omega$. I claim that $\text{Cl}_n(\text{Cl}_m(R)) = \text{Cl}_{n+m}(R)$.

Proof: I induct on n . For $n = 0$, $\text{Cl}_0(\text{Cl}_m(R)) = \text{Cl}_m(R) = \text{Cl}_{0+m}(R)$. Sps the result for n .

$$\text{Cl}_{n+1}(\text{Cl}_m(R)) = \bigcup_{q \in \text{Cl}_n(\text{Cl}_m(R))} \delta(q, \epsilon) = \bigcup_{q \in \text{Cl}_{n+m}(R)} \delta(q, \epsilon) = \text{Cl}_{n+m+1}(R).$$

I claim that $\text{Cl}(\text{Cl}(R)) = R$.

Proof: It is given $\text{Cl}(\text{Cl}(R)) \supset \text{Cl}(R)$. Suppose $q \in \text{Cl}(\text{Cl}(R))$, so $q \in \text{Cl}_n(\text{Cl}(R))$ for some $n < \omega$. Since $\text{Cl}(R) = \bigcup_{m < \omega} \text{Cl}_m(R)$, we have $\text{Cl}_n(\text{Cl}(R)) = \bigcup_{m < \omega} \text{Cl}_n(\text{Cl}_m(R))$, so $q \in \text{Cl}_n(\text{Cl}_m(R)) = \text{Cl}_{n+m}(R)$ for some $n, m < \omega$. It follows that $q \in \text{Cl}(R)$.

I claim that if $R \subset S$, then $\text{Cl}(R) \subset \text{Cl}(S)$.

Proof: Write $\text{Cl}_n(S) = \text{Cl}_n(S - R) \cup \text{Cl}_n(R)$, so $\text{Cl}_n(S) \supset \text{Cl}_n(R)$. Taking unions, we have the result.

I claim that $\text{Cl}(R) = \bigcup_{i \in I} \text{Cl}(R_i)$.

Proof: We have $\text{Cl}(R_i) \subset \text{Cl}(R)$ since $R_i \subset R$. Unioning over $i \in I$ gives one direction. Suppose $q \in \text{Cl}(R)$. Then $q \in \text{Cl}_n(R)$ for some $n < \omega$. Since $\text{Cl}_n(R) = \bigcup_{i \in I} \text{Cl}_n(R_i)$, we have $q \in \text{Cl}_n(R_i)$ for some $i \in I$, so $q \in \text{Cl}(R_i)$.

Define $\hat{\delta}$ inductively as follows. Let $\hat{\delta}(R, \epsilon) = \text{Cl}(R)$. Let $\hat{\delta}(R, xa) = \text{Cl}\left(\bigcup_{r \in \hat{\delta}(R, x)} \delta(r, a)\right)$.

I claim that $\hat{\delta}(R, w)$ is closed for all $w \in \Sigma^*$.

Proof: I induct on w . For $\hat{\delta}(R, \epsilon) = \text{Cl}(R)$ is closed. Sps the result for x . Then

$$\hat{\delta}(R, xa) = \text{Cl}\left(\bigcup_{q \in \hat{\delta}(R, x)} \delta(q, a)\right)$$

is closed.

I claim for all $x, y \in \Sigma^*$ that $\hat{\delta}(R, xy) = \hat{\delta}(\hat{\delta}(R, x), y)$.

Proof: I induct on y .

$$\begin{aligned}\hat{\delta}(\hat{\delta}(R, x), \epsilon) &= \text{Cl} \left(\bigcup_{r \in \hat{\delta}(R, x)} \delta(r, \epsilon) \right) = \text{Cl} \left(\bigcup_{r \in \hat{\delta}(R, x)} \{r\} \right) \\ &= \text{Cl}(\hat{\delta}(R, x)) = \hat{\delta}(R, x) = \hat{\delta}(R, x\epsilon).\end{aligned}$$

Sps the result for y . Then

$$\begin{aligned}\hat{\delta}(\hat{\delta}(R, x), ya) &= \text{Cl} \left(\bigcup_{r \in \hat{\delta}(\hat{\delta}(R, x), y)} \delta(r, a) \right) = \text{Cl} \left(\bigcup_{r \in \hat{\delta}(R, xy)} \delta(r, a) \right) \\ &= \hat{\delta}(R, xya),\end{aligned}$$

completing the induction (associativity of concatenation was used).

Let $E = (Q_E, \Sigma, \delta_E, q_0, F_E)$ be an ϵ -NFA. Suppose $S \subset Q_E$ and define $\hat{\delta}_E(S, \epsilon) = \text{Cl}(S)$ and inductively $\hat{\delta}_E(S, xa) = \text{Cl} \left(\bigcup_{q \in \hat{\delta}_E(S, x)} \delta_E(q, a) \right)$. Let $L(E) = \{w \in \Sigma^* : \hat{\delta}_E(q_0, w) \cap F_E \neq \emptyset\}$.

Let $D = (Q_D, \Sigma, \delta_D, q_D, F_D)$ be a DFA such that $Q_D = \{S \subset Q_E : \text{Cl}(S) = S\}$, $\delta_D : Q_D \times \Sigma \rightarrow Q_D$ is given by $\delta_D(S, a) = \text{Cl} \left(\bigcup_{q \in S} \delta_E(q, a) \right)$, $F_D = \{S \in Q_D : S \cap F_E \neq \emptyset\}$. Let $q_D = \text{Cl}(q_0)$. Note that indeed $q_D \in Q_D$ and δ_D is well-defined.

I claim that $\hat{\delta}_D(\text{Cl}(S), w) = \hat{\delta}_E(S, w)$ for all $S \subset Q_E$. When $w = \epsilon$, $\hat{\delta}_D(\text{Cl}(S), \epsilon) = \text{Cl}(S) = \hat{\delta}_E(S, \epsilon)$. Suppose the result for the string x . Then

$$\begin{aligned}\hat{\delta}_D(\text{Cl}(S), xa) &= \delta_D(\hat{\delta}_D(\text{Cl}(S), x), a) = \delta_D(\hat{\delta}_E(S, x), a) \\ &= \text{Cl} \left(\bigcup_{q \in \hat{\delta}_E(S, x)} \delta_E(q, a) \right) = \hat{\delta}_E(S, xa),\end{aligned}$$

as desired.

Suppose $w \in L(D)$, so $\hat{\delta}_D(q_D, w) \in F_D$, ie $\hat{\delta}_D(q_D, w) \cap F_E \neq \emptyset$. Since $\hat{\delta}_D(q_D, w) = \hat{\delta}_E(q_0, w)$, we have $\hat{\delta}_E(q_0, w) \cap F_E \neq \emptyset$, so $w \in L(E)$. The converse is similar.

3 Turing Machines

Let $\mathcal{P}$ denote the set of programs, so $\mathcal{P} \subseteq \{0,1\}^*$. Sps FTSOC there exists $H_0 : \mathcal{P} \times \{0,1\}^* \rightarrow \{\text{yes}, \text{no}\}$ such that $H_0(P, I) = \text{yes}$ if $P(I)$ is good and $H_0(P, I) = \text{no}$ otherwise. Let $H_1 : \mathcal{P} \times \{0,1\}^* \rightarrow \{\text{yes}, \text{good}\}$ be defined by $H_1(P, I) = \text{yes}$ and $H_0(P, I) = \text{yes}$ and $H_1(P, I) = \text{good}$ if $H_0(P, I) = \text{no}$. Define $H_2 : \mathcal{P} \rightarrow \{\text{yes}, \text{good}\}$ by $H_2(P) = H_1(P, P)$.

Sps $H_2(H_2) = H_1(H_2, H_2) = \text{yes}$. Then $H_0(H_2, H_2) = \text{yes}$, so $H_2(H_2)$ is good. Since $\text{im } H_2 = \{\text{yes}, \text{good}\}$, we have in fact $H_2(H_2) = \text{good}$, but $\text{yes} \neq \text{good}$, absurd.

Sps $H_2(H_2) = H_1(H_2, H_2) = \text{good}$, so $H_0(H_2, H_2) = \text{no}$. However, $H_2(H_2)$ is good, so $H_0(H_2, H_2) = \text{yes}$, absurd.

Let R be a TM such that if T is a TM and $s \in \Sigma^*$, then $R(\langle T, w \rangle) = 1$ if T halts on w and $R(\langle T, w \rangle) = 0$ otherwise. Let $R_1(\langle T \rangle) = R(\langle T, \langle T \rangle \rangle)$. Let $R_1^{opp} = 1 - R_1$.

Consider $R_1^{opp}(\langle R_1^{opp} \rangle) = 1$. Then $R_1(\langle R_1^{opp} \rangle) = 0$, so $R(\langle R_1^{opp}, \langle R_1^{opp} \rangle \rangle) = 0$, so R_1^{opp} doesn't halt on $\langle R_1^{opp} \rangle$, absurd. Now suppose $R_1^{opp}(\langle R_1^{opp} \rangle) = 0$. Then $R_1(\langle R_1^{opp} \rangle) = 1$, so R_1^{opp} halts on $\langle R_1^{opp} \rangle$.

4 The Universal Language

The set of strings Σ^* in the finite alphabet $\{0,1\}$ is countable. Let $\{w_i : i \in \mathbb{Z}^+\}$ be an enumeration of Σ^* . If w_i is the encoding of a TM T_i , let T_i be the associated TM. Else, let T_i be the TM of one node and no transitions (so $L(T_i) = \emptyset$).

Claim: Define $L_h = \{w_i 111w_i : T_i \text{ halts on } w_i\}$. Then L_h is undecidable.

Proof: Suppose FTSOC that L_h is decidable, so there exists a RTM M such that $L(M) = L_h$. Let M' be a TM such that M' halts on w_i iff $w_i 111w_i \notin L(M)$. M' exists since we copy $w_i \mapsto w_i 111w_i$, run M , and add infinite loops to the accepting states of M .

Let $w_i = \langle M' \rangle$. Consider M' on w_i . Suppose M' halts on w_i . Then $w_i 111w_i \notin L(M)$, which is equivalent to M' not halting on w_i , absurd. Suppose M' doesn't halt on w_i . Then $w_i 111w_i \in L(M)$, so M' halts on w_i , absurd.

Claim: Let M be a decider. Then there exists an equivalent decider M' such that M' has unique accepting and rejecting states.

Proof: Let $M = (Q, \Gamma, \Sigma, \delta, q_0, F)$. Let $Q' = Q \cup \{q', q''\}$, $q'_0 = q_0$. The unique rejecting, accepting states of M' are q', q'' . For every $q \in F$, in δ' remove all

out transitions from q and allow all transitions to q'' . For every invalid move (q, a) in δ , add a transition (q, a) to q' . Then M' is equivalent to M .

Ackermann's function. Define $A : N \times N \rightarrow N$ by the following: $A(0, y) = 1$ for $y \geq 0$, $A(1, 0) = 2$, $A(x, 0) = x + 2$ for $x \geq 2$. $A(x + 1, y + 1) = A(A(x, y + 1), y)$.

(1). $A(2, 1) = A(A(1, 1), 0)$. $A(1, 1) = A(A(0, 1), 0) = A(1, 0) = 2$. $A(2, 1) = A(2, 0) = 4$.

(2). $A(0, 1) = 1, A(1, 1) = 2, A(2, 1) = A(A(1, 1), 0) = A(1, 1) + 2 = 4$. $A(3, 1) = A(A(2, 1), 0) = A(4, 0) = 6$. I claim that $A(x, 1) = 2x$ for $x \geq 1$. Indeed, $A(1, 1) = 2$. Suppose $A(x, 1) = 2x$. Then $A(x + 1, 1) = A(A(x, 1), 0) = A(x, 1) + 2 = 2x + 2 = 2(x + 1)$, completing the induction. Then $A(0, 2) = 1$, $A(1, 2) = A(A(0, 2), 1) = A(1, 1) = 2$. $A(2, 2) = A(A(1, 2), 1) = A(2, 1) = 4$. $A(3, 2) = A(A(2, 2), 1) = A(4, 1) = 8$. I claim that $A(n, 2) = 2^n$ for all $n \geq 0$. The base case is $A(0, 2) = 1$. Assume the result for n . Then $A(n + 1, 2) = A(A(n, 2), 1) = A(2^n, 1) = 2 \cdot 2^n = 2^{n+1}$, since $2^n > 0$.

(3). Then $A(0, 3) = 1$. $A(1, 3) = A(A(0, 3), 2) = A(1, 2) = 2$. $A(2, 3) = A(A(1, 3), 2) = A(2, 2) = 4$. $A(3, 3) = A(A(2, 3), 2) = A(4, 2) = 2^4 = 16$. $A(4, 3) = A(A(3, 3), 2) = A(16, 2) = 2^{16} = 2^{2^{2^2}}$. In general, $A(n, 3) = 2^{2^{2^{\dots^2}}}$, where there are n 2's.

Claim: Let $\{w_i : i \in \mathbb{Z}^+\}$ be an enumeration of Σ^* in numerical order. $L \leq \Sigma^*$ is recursive iff L has an enumeration (by a TM) respecting this order.

Proof: Suppose L is recursive. For each $i \in \mathbb{Z}^+$, run the TM M accepting L on w_i . Call this TM M' . If $w_i \in L$, then write 10^i to the output. Since M is a decider, M' is a valid TM (since M' doesn't get stuck on some i).

Now suppose an M' enumerating L in order exists. Let $w \in \Sigma^*$. Output the M' until the last word checked by M' is greater than w . If the index associated to w appeared, then $w \in L$. Else, $w \notin L$. Since w has a finite index, the associated M is a decider.

Claim: Let $\{L_i : i \in [n]\}$ be a partition of Σ^* into RE languages. Then each L_i is recursive.

Proof: Let $L'_j = \bigcup_{i \in [j]} L_i$ for $j \in [n - 1]$. RE languages are closed under finite union, so L'_j is RE for each j . In particular, L'_{n-1} is RE. Note that $L'_{n-1} = \overline{L_n}$, so L'_{n-1}, L_n are recursive. Union is commutative, so L_i is recursive for each $i \in [n]$ (the indexing of the L_i 's is arbitrary).

Claim: Let $\{L_\alpha : \alpha \in J\}$ be an arbitrary collection of RE languages. Then $L = \bigcup_{\alpha \in J} L_\alpha$ is RE.

Proof: The set of TM's (up to equivalence) is countable. Thus, the set of RE languages is countable. The indexing J in L can be replaced by a countable $J' \leq J$. If J' is finite, we're done.

Else, let $T_1, T_2, \dots$ be a (countable) enumeration of the TM's accepting the L_i 's. Run the first i steps of the first i TM's and accept if one of the T_j reaches an accepting state (for $j \in [i]$), and repeat as $i \in \mathbb{Z}^+$.

5 Undecidable Problems about TM's

Claim: Suppose there exists a reduction from P_1 to P_2 , ie there exists a decider T that converts an instance w of P_1 to an instance of P_2 such that $w \in P_1$ iff $w \in P_2$. Then: If P_1 is undecidable, so is P_2 . If P_1 is not RE, then so is P_2 .

Proof: Suppose P_2 is decidable. Let M be a decider of P_2 . Then M' , a decider of P_1 is constructed as follows: (1). Apply T to input w . (2). Apply M to the result of T . Hence, P_1 is decidable. The contrapositive gives the desired result.

Suppose P_2 is RE. The construction is the same. Applying T does not cause the program to run loop infinitely. Then M can be applied, possibly looping infinitely. Either way, M' accepts P_1 .

Claim: Define $L_e = \{M : L(M) = \emptyset\}$ and $L_{ne} = \{M : L(M) \neq \emptyset\}$, so L_e, L_{ne} partition Σ^* . Then L_{ne} is RE, but not recursive, and L_e is not recursive.

Proof: Let M be a TM. Enumerate Σ^* in order. If for $w \in \Sigma^*$ such that w is accepted by M , we're done. If $L(M) \neq \emptyset$, then there exists $w \in L(M)$. In particular, there is an index for which we find that $M \in L_{ne}$. Hence, L_{ne} is RE.

I claim that L_{ne} is undecidable. Let $w \in \Sigma^*$. If w is of the form $w_1 111 w_2$, where w_1 represents a TM, let $w \mapsto M$, where M is the TM accepting w_2 only whose base TM is equivalent to that of w_1 . Else, let $w \mapsto M$, where M is the trivial TM.

If $\langle M, w \rangle \in L_u$, then $\langle M, w \rangle \mapsto M'$, where M' has the construction for M from before. Then $M' \in L_{ne}$. If $\langle M, w \rangle \notin L_u$, then $M' \notin L_{ne}$. Else, the input is invalid, and is mapped to the trivial TM M_0 , and $M_0 \notin L_{ne}$. Hence, L_u reduces to L_{ne} , so L_{ne} is undecidable.

Since L_e, L_{ne} are complements and L_{ne} is not recursive, L_e is not RE.

I claim that L_d can be reduced to L_e . The construction is similar to before. Let T_i be the TM associated to w_i . Let T'_i be T_i restricted to the input w_i . Let $w_i \mapsto \langle T'_i \rangle$. If $w_i \in L_d$, then $w_i \notin L(T_i)$, so $L(T'_i) = \emptyset$, so $\langle T'_i \rangle \in L_e$. If $w_i \notin L_d$, then $w_i \in L(T_i)$, so $L(T'_i) \neq \emptyset$, so $\langle T'_i \rangle \notin L_e$. Since L_d is not RE, so is L_e .

Claim: Every nontrivial property P of the RE languages is undecidable.

Proof: Assume $\emptyset \notin P$. Let $L \in P$ be nonempty and M_L the TM accepting L . Given the TM-word pair (M, w) , let M' be the TM by running M on w (say by the UTM). If $w \in L(M)$, then run M_L on the input x to M' . Else, do nothing, so $L(M') = \emptyset$. Hence, if $\langle M, w \rangle \in L_u$, then $M' \in P$. If $\langle M, w \rangle \notin L_u$, then $M' \notin P$, so L_u reduces to P .

9.3.4. (a). Yes. (b). The property P : “Is $L(M)$ finite?” is RE since Σ^* can be enumerated in dictionary order. $\emptyset$ is RE finite and Σ^* is RE infinite, so by Rice’s theorem P is undecidable. It follows that the complement property $\bar{P}$: “Is $L(M)$ infinite?” is not RE.

6 HW 4 Review

1.1. Show that $L = \{a^n b^{n^2} : n > 0\}$ is not CF. Sps FTSOC L is CF. Let N be the param of L . Let $w = a^N b^{N^2}$, so $|w| \geq N$, and $w = uvwxy$ st $|vwx| \leq N$ and $vx \neq \epsilon$. Pump up, so let $w' = uv^2wx^2y$, so $|w'| = |w| + |vx| \leq N + N^2 + N = N^2 + 2N$. Also, $|w'| > |w| = N + N^2$ since $vx \neq \epsilon$. Note that $(N + 1) + (N + 1)^2 = 2 + 3N + N^2 > 2N + N^2$, so $w' \notin L$.